

Data Handling and Infrastructure for AI

Lecture 1: Introduction

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SETU

2026



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Experience:

- Building AI systems since 2012
- PhD in 2018
- Built models for audio, text, images, video, code and more
- Pre-trained foundation models using 512 GPUs and 8 TB of data

Abdul Wahid

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Experience:

- PhD in 2023
- Research topics: LLMs, knowledge graphs, semantic technologies and AI governance
- I collaborate with Fidelity Investments through a Research Ireland Industry Fellowship

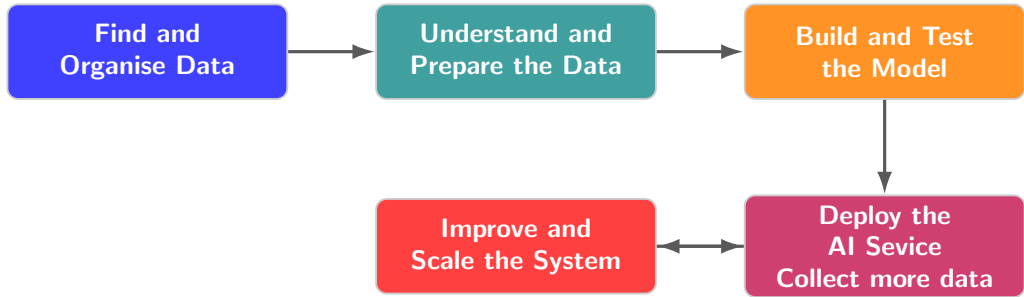
What is this course about?

- Modern AI/ML systems depend on much more than machine learning algorithms.
- Data must be collected, stored, processed, validated, and made available efficiently.
- AI/ML workloads require scalable infrastructure and distributed computing.
- This course connects **data engineering**, **machine learning operations**, and **high-performance/cloud computing**.

Main goal

Understand the infrastructure and data-handling techniques needed to build reliable, scalable AI/ML systems.

From Data to Operational AI



Our journey through the module

From raw data to a reliable, scalable and usable AI system

Course Schedule

2026 Semester 1		S	M	T	W	T	F	S
1	September 1		31	1	2	3	4	5
	2	6	7	8	9	10	11	12
	3	13	14	15	16	17	18	19
2	October 4	20	21	22	23	24	25	26
	5	27	28	29	30	1	2	3
	6	4	5	6	7	8	9	10
3	November 7	11	12	13	14	15	16	17
	8	18	19	20	21	22	23	24
	9	25	26	27	28	29	30	31
4	December 10	1	2	3	4	5	6	7
	11	8	9	10	11	12	13	14
	12	15	16	17	18	19	20	21
5	Week 10	22	23	24	25	26	27	28
	Week 11	29	30	1	2	3	4	5
	Week 12	6	7	8	9	10	11	12
6	Week 13	13	14	15	16	17	18	19
	Week 14	20	21	22	23	24	25	26
	Week 15	27	28	29	30	31		

Reading Week = End of Sprint (milestone & review)

Main topics

1



Data types and storage

2



EDA and data pipelines

3



MLOps and
high-performance
computing

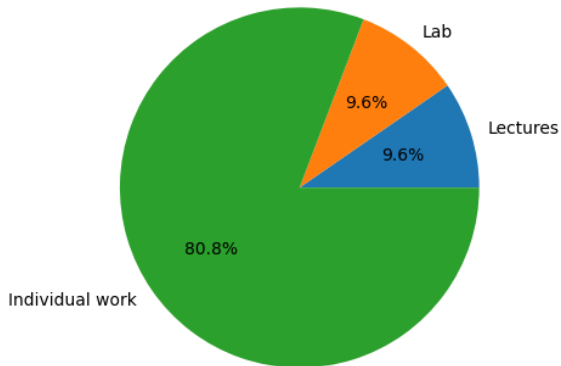
4



Scalable cloud
computing

Workload

10 ECTS = **250** - 300 work hours



1. Data Types and Storage

- Structured, semi-structured, and unstructured data
- Tabular data, text, images, audio, and video
- File formats and serialization
- Relational and NoSQL databases
- Object storage and distributed file systems
- Choosing the right storage solution

2. EDA and Data Pipelines

- Exploratory Data Analysis (EDA)
- Data cleaning and validation
- Feature preparation and transformation
- Batch versus streaming pipelines
- Data quality and reproducibility
- Automating and monitoring data workflows

3. MLOps and High-Performance Computing

- From experiments to production AI systems
- Versioning data, models, and code
- Training and inference infrastructure
- GPUs and parallel computing
- Distributed machine learning
- Monitoring, reproducibility, and deployment

4. Scalable Cloud Computing

- Cloud infrastructure for AI
- Compute, storage, and networking
- Containers and orchestration
- Elastic and distributed architectures
- Scaling AI workloads
- Cost, reliability, and performance considerations

GCP Teaching & Learning Credits

Each student will receive up to **\$50 USD in Google Cloud credits** for use during the course.

The credits can be used to experiment with cloud-based AI infrastructure, including:

- **Dataflow**: scalable data processing and pipelines
- **Scalable AI products**: training, deployment, and inference
- Cloud storage and compute resources needed for the course project

Use the credits wisely!

Cloud resources are metered. Stop or delete resources when you are finished experimenting to avoid unnecessary usage.

Using Cloud Resources Safely

Manage Your Costs

- Check your remaining cloud credits regularly.
- Use only the resources required for each task.
- Stop or delete resources when you finish.
- Avoid leaving idle virtual machines running.

Keep Your Work Secure

- Never share passwords, API keys or access tokens.
- Do not upload credentials to GitHub.
- Use approved, public or non-sensitive datasets.
- Check repository permissions before sharing.

Don't forget to check, stop and clean up your resources. (no extra GCP credits)

Academic Integrity and the Use of AI

You May Use AI Tools to Support Learning

- Explore concepts and ask for explanations.
- Debug code and examine alternative solutions.
- Improve the clarity and structure of your work.

You Remain Responsible for Your Work

- Understand and be able to explain all submitted work.
- Verify generated code, claims, results and references.
- Clearly acknowledge any use of generative AI.
- Do not submit fabricated results, citations or evidence.

AI can assist your work, but it cannot replace your understanding.

Course Requirements

Weekly quizzes/tasks — 10%

- A short quiz, activity or discussion will take place each week.
- These collectively account for **10%** of the overall score.

Project work — 90%

The course project is developed through four milestones (first 3 peer-graded):

- Milestone 1: **10%**
- Milestone 2: **10%**
- Milestone 3: **10%**
- Final report and code/system: **60%**

Milestone 1

10%

Initial idea,
problem definition,
and planning

Milestone 2

10%

Data and
infrastructure
design

Milestone 3

10%

Implementation
and intermediate
results

Final

60%

Final report +
code/system

The milestones build progressively toward a complete AI infrastructure project.



Questions?